SAFETY AND COMPLIANCE TESTING FOR YONGKANG YAODONG TECHNOLOGY CO., LTD.

Tested Sample(s)	: E-Motorcycle
Brand	: TERRAWULF/ SYKERUNS*
Model	: M7/ M7S*/ X7*
Color	: Black/ Red*/ Yellow*/ Silver*
Size	: Not Specified
Stock / Model Number	: M7/ M7S*/ X7*
Country of Origin	: China
Age Grading	: Not Specified
Children's Product	: No

**Results applicable to the brands, models and colors not received*

Prepared For:

Yongkang Yaodong Technology Co., Ltd.
Xizhuyuan Industrial Zone, Dongcheng Street,
Yongkang City, Zhejiang Province, CN

Part of the Cotecna Group

Issue Date: 10 July 2026

Final Report: 2489.22776.001

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IAS ACCREDITED Testing Laboratory

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer joint ISO-ILAC-IAF Communiqué dated April 2017.) The Joint Communiqué is available on publications and resources page of the ILAC website at http://www.ilac.org. Accreditation listing and certificate can be found at http://www.iasonline.org.

DATE AND PLACE OF TEST

Sample(s) received on : 14 May 2026
Testing was initiated on : 14 May 2026
Testing was completed on : 02 July 2026
Testing was performed at : Taicang ACT Sporting Goods Testing Co., Ltd.
No. 35 Zhenghe Road, Ludu Town,
Taicang City, Suzhou, Jiangsu Province, China 215412

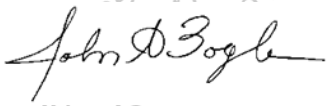
OUTSIDE LAB TESTING

Testing was performed at : STQ Testing Services Co., Ltd.
2-1-B, China-singapore science industrial zone I, 8# zhanye road, sip suzhou, China

CONCLUSION

2489.22776.001 – Yaodong, M7 (Black)		
Purpose of Test – Each test performed is intended to check compliance with the following:	**Result**	**Comment**
ANSI/CAN/UL 2272:2024 Electrical Systems for Personal E-Mobility Devices	C	

President,



John A. Bogler

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SAMPLE IDENTIFICATION

Brand:	TERRAWULF	**Job No.:**	2489.22776
Model:	M7	**Sample ID:**	2489.22776.001
Manufacturer:	Yongkang Yaodong Technology Co., Ltd.	**Type:**	E-Motorcycle
Stock/Model No.:	M7	**Material:**	Not Specified
UPC:	Not Specified	**Size:**	Not Specified
Serial No.:	Not Specified	**Color(s):**	Black
Age Grading:	Not Specified	**Weight (kg):**	51.7
Children's Product:	No	**Country of Origin:**	China



2489.22776.001 – M7 (Black)

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Test results apply to the following brands, models and colors:

Brand	Model	Color
TERRAWULF	M7S	Black
		Red
		Yellow
SYKERUNS	X7	Silver



M7S (Black)



M7S (Red)



M7S (Yellow)



X7 (Silver)

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Battery:			
Brand:	Not Specified	**Normal Voltage:**	60V
Model:	60V 27AH 1620Wh	**Rated Capacity:**	27Ah 1620Wh



Battery

Charger:			
Brand:	Zhejiang Fudian Intelligent Technology Co., Ltd.	**Input:**	100-240V 50/60Hz 5A
Model:	FY3296725000	**Output:**	67.2V 5A



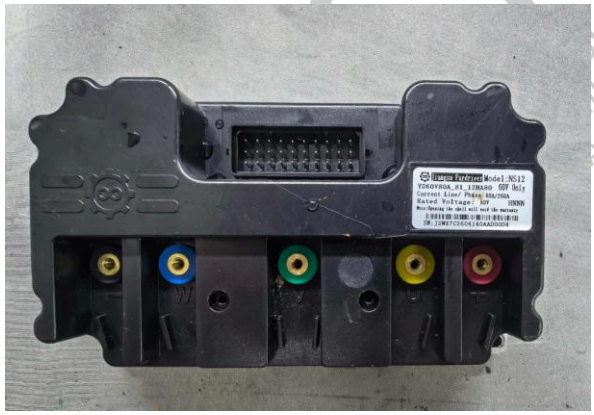
Charger

Drive Unit (Motor):			
Brand:	Not Specified	**Rated Continuous Output:**	Not Specified
Model:	YLV6W120G70	**Rated Voltage:**	60V



Drive Unit: Motor

Motor Controller:					
Brand:	Jiangsu Fardriver	**Model:**	NS12	**Serial #:**	JSWX7C2604140AAD00D4



Motor Controller (Hub Controller)

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HDI/HMI Controller:					
Brand:	Not Specified	**Model:**	G518	**Serial #:**	1008121251222110





HDI/HMI Controller

HDI/HMI Controller 2:					
Brand:	SCIWIL	**Model:**	Not Specified	**Serial #:**	Not Specified





HDI/HMI Controller

TEST METHODS

Method for each test conducted is as follows:

- UL 2272 testing was performed utilizing the test methods from ANSI/CAN/UL 2272:2024 Electrical Systems for Personal E-Mobility Devices.

TEST RESULTS

C: Compliant; Product meets specified standard **PC**: Compliant to tested sections **NC:** Non-Compliant; Product does not meet specified standard **NA:** Not Applicable to this design **NR:** Not Requested by the Applicant **NP:** Not Present	**ND:** None Detected **IC:** Inconclusive **NT:** Not Tested **FTR:** Further Testing Recommended **PPM:** Parts Per Million ***:** See Comments †: Testing performed at outside lab

ANSI/CAN/UL 2272:2024 Electrical Systems for Personal E-Mobility Devices

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
CONSTRUCTION			
7	**Non-Metallic Materials** *Note:* This section does not need to be tested if UL 2849 has been tested.		
7.1	Enclosure Materials Comply with UL746C, Path III of Enclosure Requirements in Table 4.1 (or CAN/CSA-C22.2 No. 0.17)	C	
7.2	Polymeric Materials – Minimum Flame Rating of 94V-1 (UL 94 or CAN/CSA-C22.2 No. 017) *Exception: The enclosure may alternatively be evaluated to the 20 mm end product flame test in accordance with the Standard for Polymeric Materials - Use in Electrical Equipment Evaluations,UL 746G, or the Standard for Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.17.*	C	
7.3	Resistance to impact, crush resistance, abnormal operations, sever conditions, mold-stress relief distortion	C	
7.4	Polymeric Materials – Enclosure w/ Insulation shall have Relative Thermal Index ≥ 80°C (176°F) (UL 746B or CAN/CSA-C22.2 No. 017)	C	
7.5	Enclosure Materials Exposed to Sun/Rain Meet UV Resistance and Water Exposure/Immersion Tests (UL 746C or CAN/CSA-C22.2 No. 017) This requirement may be waived if the device is marked that it is to be stored indoors and user instructions also indicate that it not be left outdoors to be exposed to UV or rain. See 45.11 and Section 46	NA	
7.6	Electrical Insulation shall be resistant to deterioration. Materials employed for direct support of live parts at hazardous voltage, shall additionally meet the direct support insulation criteria (UL 746C or AN/CSA-C22.2 No. 0.17), unless employed as part of a component that has been evaluated to a suitable component standard. Insulated wiring is subjected to the requirements outlined in Section 10, Wiring and Terminals.	C	
7.7	Gaskets and seals relied upon for safety must be tested according to **one of the following** standards: a) UL 157 b) UL 62368-1 / CSA C22.2 No. 62368-1 * c) UL 50E / CSA C22.2 No. 94.2 *	NA	

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
	* Only Gasket Tests from those standards		
7.8	Nonmetallic materials for internal parts must be rated V-2 minimum. *Exception:* *for PS2 circuits (UL 62368-1 / CSA C22.2 No. 62368-1)* *a) Mounted on minimum V-1 or VTM-1 class material.* *b) Constructed of minimum V-2, VTM-2, or HF-2 class foamed material.* *c) Size less than 1750 mm³ (0.11 in³).* *d) Mass of combustible material less than 4 g (0.14 oz).* *e) Separated by at least 13 mm (0.51 in) of air from electrical parts likely to produce high temperatures under fault conditions.* f) *Enclosed in a sealed enclosure of 0.06 m³ (2.12 ft³) or less, made of non-combustible material with no ventilation openings.*	NA	
7.9	Internal parts of components must comply with the flammability requirements specified in Section 2 of the component standard.	C	
7.10	Small parts and gaskets not near live parts and unable to propagate flame within the equipment do not require a specific flame rating.		C
7.11	Nonmetallic materials outside the enclosure and not part of it are decorative and do not require a specific flame rating.		NA
7.12	Printed wiring board materials used for circuits or components at hazardous voltage or hazardous energy levels must be rated V-1 minimum.	NA	
7.13	For the requirements outlined in 7.8 – 7.12, the flammability rating of the material shall be provided as part of the material rating or the flammability rating may be determined in accordance with UL 94 and CSA C22.2 No. 0.17.	C	
8	**Metallic Parts Resistance to Corrosion** *Note:* This section does not need to be tested if UL 2849 has been tested.		
8.1	Metal Enclosures – Corrosion Resistant (UL 50E or CAN/CSA-C22.2 No. 94.2)	C	
8.2	Insulation of Metal Enclosures – Non-Moisture Absorbent Materials w/ Suitable Temperature Rating.	C	
8.3	Conductive parts at terminals and connections shall not be subject to corrosion due to electrochemical action.	C	
9	**Enclosures**		
9.1	General *Note:* This section does not need to be tested if UL 2849 has been tested.		
9.1.1	The enclosure relied upon for safety per 6.14 shall have the strength and rigidity required to resist the possible physical abuses that it will be exposed to during its intended use.	C	
9.1.2	A tool providing the mechanical advantage of a pliers, screwdriver, hacksaw, or similar tool, shall be the minimum mechanical capability required to open the enclosure.	C	
9.1.3	Openings in the enclosure shall be designed to prevent inadvertent access to hazards such as hazardous voltages, hazardous moving parts or hot surfaces that could result in burns. Parts that can be removed without the use of a tool, are to be removed to determine compliance.	C	
9.1.4	Openings in the enclosure shall be designed to prevent ingress of water (minimum rating of IPX4)	C	
9.2	Battery Compartments		
9.2.1	Cell vents shall not be obstructed in such a way as to defeat their operation if venting is relied upon for compliance with this standard. Compliance is checked by inspection.	NA	
9.2.2	Battery compartments within enclosures shall secure the battery in place sufficiently to prevent excessive movement and stress on the battery and cells that could result in a hazard. Compliance is checked by inspection and the mechanical tests of this standard.	C	

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
9.2.3	The battery enclosure must prevent user servicing with simple tools and should be ultrasonically welded or secured by adhesives (UL 746C), tamper-proof screws, etc. *Exception: Can be replaced by detectable openings (e.g., broken seal). Users must be reminded not to use products with broken seals and to recycle them.*	NA	
9.2.4	The device enclosure serving as the battery's outer enclosure must prevent user servicing with common tools. It should be ultrasonically welded or secured by adhesives (UL 746C), or tamper-proof screws. Reference: See 47.12 and 48.4	NA	
10	**Wiring And Terminals** *Note:* This section does not need to be tested if UL 2849 has been tested.		
10.1	Wiring shall be insulated properly	C	
10.2	Internal wiring must be secured to prevent strain, terminal loosening, and insulation damage. In safety-critical circuits, soldered terminations should not rely solely on solder, and wire routing must not stress battery cells in assembled products.	C	
10.3	Connections to cells must avoid damaging them. High-heat processes like soldering should not be used directly on cell terminals without proper controls to prevent heat damage.	C	
10.4	External Terminals – designed to prevent inadvertent shorting, misalignment, or disconnection when scooter is in use	C	
10.5	Removable Battery Packs – Terminals not readily accessible	C	
10.6	Removable Battery Packs – Endurance Test (UL 2251 or CAN/CSA C22.2 No. 282)	C	
10.7	Holes for Wiring – shall be provided with a smoothly rounded bushing or shall have a smooth surface, free of burrs, fins, sharp edges, etc.	C	
10.8	Hazardous Voltage Warning Label (ISO 7010, No. W012 – i.e., lightning bolt within triangle)	NA	
11	**Chargers** *Note:* This section does not need to be tested if UL 2849 has been tested.		
11.1	Chargers for personal e-mobility devices must meet intrinsic safety requirements as per standards UL 1310, CSA-C22.2 No. 223, UL 1012, CSA-C22.2 No. 107.1, UL 60950-1, or UL 62368-1. They must be compatible with the device's battery system, with compliance confirmed through data review and standard tests. Chargers should connect to a standard outlet if they are to be linked to a main electricity supply.	C	
11.2	The charger connector must prevent misalignment, reverse polarity, or electrical mismatch. Compliance is checked by design inspection and, if necessary, by reviewing Section 16 Protective Circuits and Safety Analysis requirements.	C	
12	**Fuses** *Note:* This section does not need to be tested if UL 2849 has been tested.		
12.1	Fuses shall be acceptable for the current and voltage of the circuit they protect.	C	
12.2	Replaceable Fuses – Replacement properly/obviously marked adjacent to holder. Markings for user-replaceable fuses must be clear, showing ratings and, if special characteristics like time delay or breaking capacity are relevant, the type. Instructions for correct fuse replacement must also be provided.	NA	
13	**Lighting**		
13.1	If provided with user replaceable bulbs, replacement does not impair the safety of the personal e-mobility device and there is no risk of electric shock. Instructions provided with the personal e-mobility device includes information on the type and ratings of user replaceable bulbs.	NA	
14	Electrical Spacings and Separation of Circuits *Note:* This section does not need to be tested if UL 2849 has been tested.	C	

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
15	**Insulation Levels and Protective Grounding**		
15.1	Hazardous Voltage Circuits – Insulated from accessible conduction parts and safety extra low voltage circuits (60 Vdc or 48 Vrms)	C	
15.4	Protective Ground System – Max Resistance of 0.1 Ω	NA	
15.5	Ground Terminal Identification: a) A green-colored, not readily removable terminal screw with a hexagonal head; b) A green-colored, hexagonal, not readily removable terminal nut; c) A green colored pressure wire connector; or d) The word "Ground" or the letters "G" or "GR" or the grounding symbol (IEC 60417, No. 5019 - upside down "tree" in circle) or otherwise identified by a distinctive green color. For Canada only use the grounding symbol rather than the word "Ground" or the letters "G" or "GR".	NA	
15.6	Conductor shall be properly sized – if insulated, the insulation shall be green or green & yellow striped in color	NA	
16	**Protective Circuits and Safety Analysis** *Note:* This section does not need to be tested if UL 2849 has been tested.		
16.2	Analysis of potential electrical and energy hazards (FMEA)	C	
16.4	Critical Safety Circuits – provided with redundant passive protection, redundant active protection, or are determined to fail safe upon loss of power to/failure of the active circuit.	C	
16.5	Electronic and Software Protection Scheme (UL 991, UL 60730-1, IEC 61508-1)	NA	
16.6	If the device contains hazardous voltages, it has a manual disconnect that: a) Disconnects both poles of the hazardous voltage circuit; b) Requires manual action to break the electrical connection; c) Ensures disconnection is physically verifiable; and, d) When engaged does not create exposed conductors and is insulted to prevent shock hazard.	C	
16.7	Automatic Disconnect Requirements: a) No auto reset; b) Disconnects both poles; c) Capable of full load disconnects; and d) No hazardous conditions upon automatic actuation.	C	
16.8	Charger connect-interlock. Unit cannot be activated when charger is plugged in	NA	
17	**Cells and Batteries** *Note:* This section does not need to be tested if UL 2849 has been tested.		
17.2	Lithium based Cells – comply w/ UL 2580	C	
17.4	Nickel Based Cells – comply w/ UL 2580	NA	
17.5	Valve regulated lead acid batteries comply with pressure release test from UL 1989	NA	
17.6	Electrochemical capacitors shall comply with the capacitor requirements in UL 810A	NA	
17.7	Sodium nickel metal chloride cells shall comply with the Sodium-Beta Battery Cells Test Program Annex in UL 1973.	NA	
17.8	Sodium ion cells shall comply with the Cell Test Program Annex in UL 1973.	NA	
17.9	Batteries complying with UL/ULC 2271 are exempt from: Overcharge Test, Short Circuit Test, Over-discharge Test, Imbalanced Charging Test.		NA
17.10	Batteries powering non-drivetrain units shall comply with UL 2054, UL 62133-2 / CSA C22.2 No. 62133-2, or UL 62133-1 / CSA C22.2 No. 62133-1.	C	
18	**Motors** *Note:* This section does not need to be tested if UL 2849 has been tested.		
18.1	Traction motor is not hazardous under locked rotor and overload conditions.	C	

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
18.2	Motors shall be capable of carrying max normal anticipated load without exceeding temperatures on insulation and windings.	C	
18.3	Motors in hazardous voltage circuits must meet UL 1004-1 or CSA-C22.2 No. 100 standards. Motors in low voltage circuits must comply with either of these standards or this standard's requirements.	C	
19	**Manufacturing and Production Line Testing**		
19.6	Continuity check of the grounding conductors – resistance of the protective grounding system ≤ 0.1 Ω	NA	
19.7	Documentation of production process	C	
	Performance (not a test)		
22	Post Test Cycle		
	Electrical Tests		
24	Overcharge Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
25	Short Circuit Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
26	Over-discharge Test	C	
27	Temperature Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
28	Post-discharge Charging Determination Test		
28.1	Immediately after the Temperature Test in Section 27, connect the charger to the DUT. Check if the unit allows charging above the cell manufacturer's max specified cell surface temperature. Ensure the charger and DUT electrical connection comply with 27.4 and 27.6. Bypassing the BMS may be required for cell measurement.	C	
29	Imbalanced Charging Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
30	Dielectric Voltage Withstand Test *Note 1:* This section does not need to be tested if UL 2849 has been tested. *Note 2*: This test is only conducted as part of the requirements of Hazardous Voltage Only testing. Refer to the Hazardous Voltage Only section for results.		
31	Isolation Resistance Test *Note 1:* This section does not need to be tested if UL 2849 has been tested. *Note 2*: This test is only conducted as part of the requirements of Hazardous Voltage Only testing. Refer to the Hazardous Voltage Only section for results.		
32	Leakage Current Test	NA	
33	Grounding Continuity Test	NA	
	Mechanical Tests		
34	Vibration Test	C†	As per STQ test report SZ2026050983
35	Shock Test	C†	
36	Input Test		
36.1	The input current of a product is to be measured while charging a fully discharged battery. The input current shall not exceed 110% of the manufacturer's rated value for the DUT. If an external charger is used, the measured input current shall not exceed the charger's rated output current.	C	
37	Crush Test	NA	
38	Drop Test	C	
39	Mold Stress Relief Test	C	

UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
40	Handle Loading Test	NA	
41	Motor Overload Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
42	Motor Locked Rotor *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
43	**Strain Relief Test (Cord Anchorages)** *Note:* This section does not need to be tested if UL 2849 has been tested.		
43.2	Strain Relief Pull Test	C	
43.3	Push-Back Test	C	
	ENVIRONMENTAL TESTS		
44	**Water Exposure Tests** *Note:* This section does not need to be tested if UL 2849 has been tested.		
44.1	IPX4 Code Rating	C	
44.2	Partial Immersion	NA	
45	Thermal Cycling Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
46	Label Permanence Test *Note:* This section does not need to be tested if UL 2849 has been tested.	C	
	MARKINGS		
47.1	Compliance markings must be legible and permanent, like etched or adhesive labels. Adhesive labels must meet UL 969 or CSA-C22.2 No. 0.15 standards for their intended use conditions. An exception allows for adhesive labels to be tested for permanence as outlined in Section 46.	C	
47.2	Mandatory Markings: Manufacturer's Name, trade name, trademark, or other identifying mark; Part # or Model #; Electric Ratings (Vdc and Ah or Wh); Max Weight (in lbs or kg); and Max Speed (in mph or km/h).	C	
47.3	Date of Manufacture or Traceable Date Code	NA	
47.4	Charging Instructions ("Use Only (__) Charger")	C	
47.5	Mark all external terminals and connections with electrical ratings unless they comply with UL 62368-1 / CSA C22.2 No. 62368-1, Annex Q.1, or the Low-Voltage, Limited-Energy Circuits table in UL 2849, including polarity markings if applicable.	NA	
47.6	Separable Battery Pack Specs – "Use only (---) battery pack w/ this scooter…"	NA	
47.7	Ground Connection Markings	NA	
47.8	Warning for Hazardous Voltage Circuits	C	
47.9	Warning about reading instruction manual.	C	
47.10	Marks for IPX4 rating not required. Scooters marked w/ higher IP ratings shall comply with those ratings.	C	
47.11	Plastic enclosure not evaluated for exposure to UV light and rain per 7.5 shall be marked with the equivalent, "Store Indoors When Not in Use."	NA	
47.12	Mark the battery or device enclosure with: 'WARNING – Risk of Fire and Electric Shock – Battery and/or battery components are not user replaceable.	C	
	INSTRUCTIONS		
48.1	Shall Include:		
	Charging Instructions	C	
	Operating Instructions	C	
	Storage and Disposal Instructions	C	

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UL 2272:2024 Electrical Systems for Personal E-Mobility Devices			
Ref. #	**Test Description**	**Result**	**Observations and Notes**
	Temperature Limits	C	
	Appropriate Charger Specs	C	
	Weight Limits (min and max)	C	
	Max Speed	C	
	Instructions for Water and Other Environmental Exposures	C	
	Instructions for Riding Surface/Terrain, Use on Gradients, etc.	C	
	Instructions for Replaceable Fuses and Light Bulbs	NA	
48.2	Removable Battery Pack Instructions	C	
48.3	Warning about Risk of Fire and Electric Shock – No User Serviceable Parts	C	
48.4	Include in the instructions: 'WARNING: Risk of Fire and Electric Shock – Battery and/or battery components are not user replaceable.'	C	
48.5	Devices not intended for use in high altitude locations shall indicate not intended for use at elevations greater than 2000 m above sea level.	NA	
48.6	Devices intended to be stored indoors to protect against prolonged exposure to UV rays or the elements that may damage enclosure materials per 7.5, shall have the following or equivalent included in the instruction manual: "Prolonged Exposure to UV Rays, Rain and the Elements May Damage the Enclosure Materials, Store Indoors When Not in Use".	C	

NOTICE

1. The report is not effective without the signature of the person(s) authorizing the report (ACT Lab's authorized signatory is John A. Bogler (President)).
2. The report is not valid if altered.
3. Claims have to be made within 15 days after receipt of this report.
4. The results of this test report relate only to the items tested.
5. The results apply to the samples as received.
6. For reports that contain results from external test service providers: Results from external test service providers are supplied by the customer and can affect validity of results.
7. Decision rule applied according to Section 4.2.1 “Binary Statement for Simple Acceptance Rule” as stated in "ILAC-G8:09/2019 - Guidelines on the Reporting of Compliance with Specification".

****

END OF REPORT

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